



电力系统稳态分析作业参考答案

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注意：本资料仅供学习参考，不作为标准答案，请不要拿本资料与老师对峙！

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第 1 章 作业

1-1

(1) 教材 P14: 然而, 在实际电力系统中有大量的输电线路和配电线路, 它们输送功率的大小和距离各不相同, 不可能, 也没有必要对它们分别采用不同的“最佳电压”。特别是从设备制造的经济性和运行维护的方便性来说, 需要对设备进行规格化和系列化, 而不宜有过多的额定电压等级。为此, 世界各国都规定一定数量的标准电压, 这些标准电压通常称为电压等级。

(2) 见学辅资料第一章。

1-2

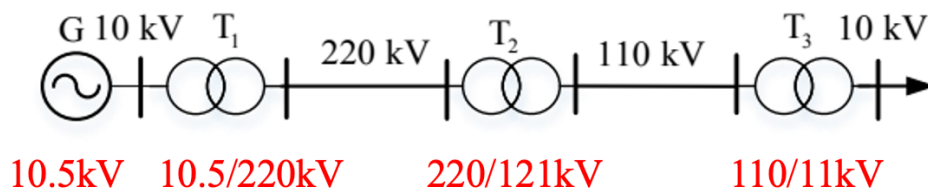


图 1.1: 习题 1-2

1-3

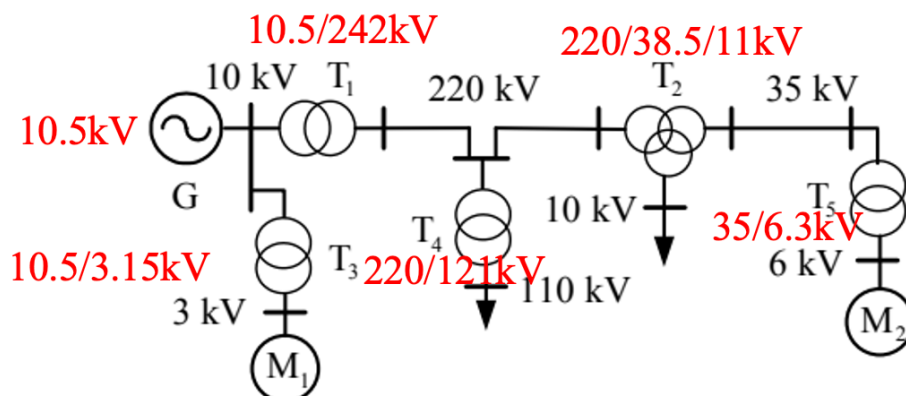


图 1.2: 习题 1-3

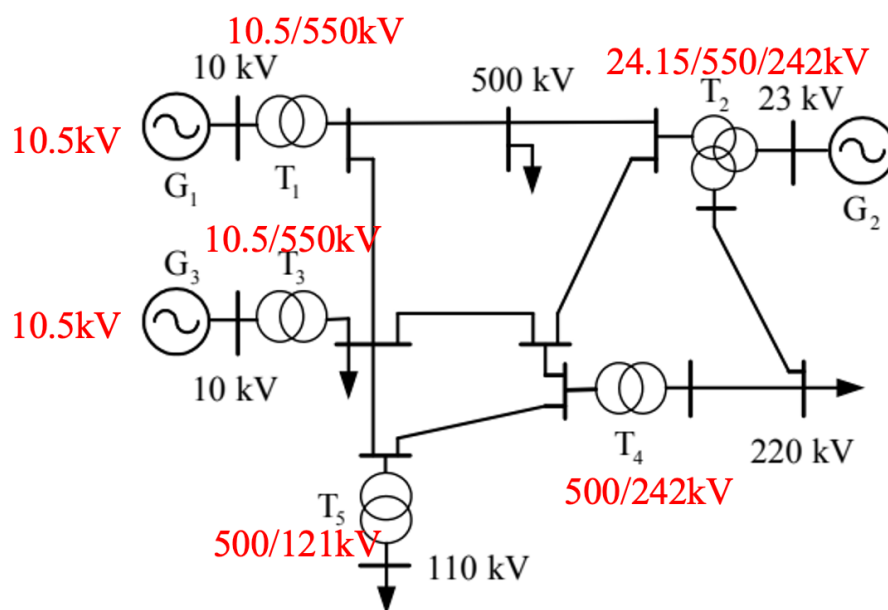


图 1.3: 习题 1-8

第 2 章 作业

2-1

注意什么条件下用集总参数，什么情况下用分布参数（教材 P39）：

$$r_1 = r_{20} [1 + \alpha(t - 20^\circ\text{C})] = 0.27 (1 + 0.0036 \times 20) = 0.289 \, \Omega/\text{km}$$

$$D_m = \sqrt[3]{3.5 \times 3.5 \times 7} = 4.4097 \, \text{m}$$

由此得单位长度参数：

$$\begin{cases} x_1 = 0.4146 \, \Omega/\text{km} \\ b_1 = 2.749 \times 10^{-6} \, \text{S}/\text{km} \end{cases}$$

则对应 100km 线路的集中参数：

$$\begin{cases} R = 28.9 \, \Omega \\ X = 41.46 \, \Omega \\ \frac{B}{2} = 1.375 \times 10^{-4} \, \text{S} \end{cases}$$

由于双回线，则阻抗为单回线的一半，导纳为两倍。

故最终参数为：

$$\begin{cases} R = 14.45 \, \Omega \\ X = 20.73 \, \Omega \\ \frac{B}{2} = 2.75 \times 10^{-4} \, \text{S} \end{cases}$$

2-2

单位长度电抗公式：

$$x_1 = 0.1445 \lg \frac{D_m}{r_{\text{eq}}} + \frac{1}{n} \cdot 0.0157 \, \mu_r$$

计算参数:

$$D_m = \sqrt[3]{12 \times 12 \times 24} = 15.12 \text{ m}, \quad r = \frac{1}{2}\phi = 13.6 \text{ mm}$$

$$r_{eq} = \sqrt[4]{13.6 \times 400 \times 400 \sqrt{2} \times 400} \text{ mm} = 187.309 \text{ mm}$$

代入得:

$$x_1 = 0.1445 \lg \frac{15.12}{187.309 \times 10^{-3}} + \frac{1}{4} \times 0.0157 = 0.279 \Omega/\text{km}$$

单位长度电纳:

$$b_1 = \frac{7.58}{\lg \frac{D_m}{r_{eq}}} \times 10^{-6} = \frac{7.58}{\lg \frac{15.12}{187.309 \times 10^{-3}}} \times 10^{-6} = 3.975 \times 10^{-6} \text{ S/km}$$

2-3

已知: 高压侧额定电压 $U_N = 110 \text{ kV}$, 额定容量 $S_N = 15000 \text{ kVA} = 15 \text{ MVA}$

变压器阻抗参数:

$$\begin{cases} R_T = \frac{P_k U_N^2}{1000 S_N^2} = \frac{133 \times 110^2}{1000 \times 15^2} = 7.152 \Omega \\ X_T = \frac{U_k \% \cdot U_N^2}{100 S_N} = \frac{10.5 \times 110^2}{100 \times 15} = 84.7 \Omega \end{cases}$$

变压器导纳参数:

$$\begin{cases} G_m = \frac{P_0}{1000 U_N^2} = \frac{50}{1000 \times 110^2} = 4.13 \times 10^{-6} \text{ S} \\ B_m = -\frac{I_0 \% \cdot S_N}{100 U_N^2} = -\frac{3.5 \times 15}{100 \times 110^2} = -4.34 \times 10^{-6} \text{ S} \end{cases}$$

2-4

没有特殊说明的话, 短路电压参数一般是折算后的:

$$\begin{cases} U_{k1} \% = \frac{1}{2}(10.5 + 17.5 - 6.5) = 10.75 \\ U_{k2} \% = \frac{1}{2}(10.5 + 6.5 - 17.5) = -0.25 \\ U_{k3} \% = \frac{1}{2}(17.5 + 6.5 - 10.5) = 6.75 \end{cases}$$

电阻参数:

$$R_{(100)} = \frac{1}{2} \cdot \frac{P_{max} U_N^2}{1000 S_N^2} = \frac{1}{2} \cdot \frac{185 \times 110^2}{1000 \times 25^2} = 1.791 \Omega$$

$$R_{(50)} = 2R_{(100)} = 3.582 \Omega$$

电抗参数:

$$\begin{cases} X_{T1} = \frac{U_{k1}\% \cdot U_N^2}{100S_N} = \frac{10.75 \times 110^2}{100 \times 25} = 52.09 \Omega \\ X_{T2} = \frac{U_{k2}\% \cdot U_N^2}{100S_N} = \frac{-0.25 \times 110^2}{100 \times 25} = -1.21 \Omega \\ X_{T3} = \frac{U_{k3}\% \cdot U_N^2}{100S_N} = \frac{6.75 \times 110^2}{100 \times 25} = 32.67 \Omega \end{cases}$$

导纳参数:

$$\begin{cases} G_m = \frac{P_0}{1000U_N^2} = \frac{52.6}{1000 \times 110^2} = 4.35 \times 10^{-6} \text{ S} \\ B_m = -\frac{I_0\% \cdot S_N}{100U_N^2} = -\frac{3.6 \times 25}{100 \times 110^2} = -7.44 \times 10^{-5} \text{ S} \end{cases}$$

2-5

三台变压器的额定容量:

$$S_N = 3 \times 3500 \text{ kVA} = 10500 \text{ kVA} = 10.5 \text{ MVA}$$

额定变比 (Y-Δ 接线):

$$k = \frac{139.72\sqrt{3}}{6.3} = 38.41$$

(注: 额定电压为线电压有效值, 高压侧 U_N 为 $139.72\sqrt{3}\text{V}$)

电阻:

$$R_T = \frac{P_k U_N^2}{1000S_N^2} = \frac{28.633 \times 3 \times (139.72\sqrt{3})^2}{1000 \times (10.5)^2} = 45.63 \Omega$$

电抗:

$$X_T = \frac{U_k\% \cdot U_N^2}{100S_N} = \frac{6.5 \times (139.72\sqrt{3})^2}{100 \times 10.5} = 342.55 \Omega$$

电导:

$$G_m = \frac{P_0}{1000U_N^2} = \frac{3 \times 10.012}{1000 \times (139.72\sqrt{3})^2} = 5.129 \times 10^{-7} \text{ S}$$

电纳:

$$B_m = -\frac{I_0\% \cdot S_N}{100U_N^2} = -\frac{1.753 \times 10.5}{100 \times (139.72\sqrt{3})^2} = 1.143 \times 10^{-6} \text{ S}$$

2-6

忽略电阻、电纳，仅保留电抗、电导。

变压器：112.75kV/6.6kV，有分接头。下列参数均为两条线路并联后的参数。

(1) 线路电抗： $X_L = \frac{1}{2} \times 0.27 \times 70 = 9.45 \Omega$

变压器参数：额定容量 $S_N = 20000 \text{ kVA} = 20 \text{ MVA}$ ，额定电压 $U_N = 110 \text{ kV}$

则电抗有名值：

$$X_T = \frac{1}{2} \times \frac{U_K\% \cdot U_N^2}{100S_N} = \frac{10.5 \times 110^2}{100 \times 20} = 31.76 \Omega$$

电抗器参数：

公式 $X_R = \frac{X_R\%}{100} \cdot \frac{U_N}{\sqrt{3}I_N}$ ，归算至 110kV 电压等级侧：

$$X_R = \frac{1}{2} \times \frac{4}{100} \cdot \frac{10}{\sqrt{3} \times 0.3} \cdot \left(\frac{112.75}{6.6} \right)^2 = 112.33 \Omega$$

电缆电抗：

$$X_D = \frac{1}{2} \times 0.08 \times 5 \times \left(\frac{112.75}{6.6} \right)^2 = 58.37 \Omega$$

(2) 标么值计算，取基准值： $S_B = 100 \text{ MVA}$ ， U_B 取额定电压等级。

线路电抗标么值：

$$X_L^* = \frac{X_L}{U_B^2/S_B} = 9.45 \times \frac{100}{110^2} = 0.078$$

变压器归算用教材 P53 的模型，归算至 110kV 电压等级侧：

变比标么值：

$$k^* = \frac{U_{2t}/U_{1t}}{U_{2B}/U_{1B}} = \frac{6.6 : 112.75}{6 : 110} = 1.073$$

低压侧电抗标么值:

$$Z_T^* = \frac{U_{K\%}}{100} \cdot \frac{U_{1N}^2}{S_N} \cdot \frac{S_B}{U_{1B}^2} = j0.263$$

代入公式计算即可，此处不再写具体计算过程：

$$Z_{TII}^* = k^* \cdot Z_T^* = j0.282$$

并联导纳标么值:

$$y_{10}^* = \frac{k^* - 1}{Z_T^* k^*} = -j0.259, \quad y_{20}^* = \frac{k^* - 1}{Z_T^* k^*} = j0.241$$

电抗器标么值:

$$X_R^* = X_R \cdot \frac{S_B}{U_B^2} = 112.33 \cdot \frac{100}{110^2} = 0.928$$

电缆线路标么值:

$$X_D^* = X_D \times \frac{S_B}{U_B^2} = 58.37 \times \frac{100}{110^2} = 0.482$$

2-7

变压器 T2:

$$\begin{aligned} U_{k1}\% &= \frac{1}{2}(15 + 20 - 10) = 12.5, & X_{T1} &= \frac{U_{k1}\%}{100} \cdot \frac{U_{1N}^2}{S_N} = 0.101 \\ U_{k2}\% &= \frac{1}{2}(15 + 10 - 20) = 2.5, & X_{T2} &= \frac{U_{k2}\%}{100} \cdot \frac{U_{2N}^2}{S_N} = 0.020 \\ U_{k3}\% &= \frac{1}{2}(10 + 20 - 15) = 7.5, & X_{T3} &= \frac{U_{k3}\%}{100} \cdot \frac{U_{3N}^2}{S_N} = 0.061 \end{aligned}$$

折算至低压侧，取 $S_B = 100\text{MVA}$ ， U_B 为对应电压等级 500kV、220kV、35kV。

$$k_{13}^* = \frac{U_{1t} : U_{3t}}{U_{1B} : U_{3B}} = \frac{550(1 + 0.025) : 24.15}{500 : 23} = 1.074$$

$$k_{23}^* = \frac{U_{2t} : U_{3t}}{U_{2B} : U_{3B}} = \frac{242(1 - 0.025) : 24.15}{220 : 23} = 1.021$$

$$Z_{T1}^* = \frac{U_{k1}\%}{100} \cdot \frac{U_{1N}^2}{S_N} \cdot \frac{S_B}{U_{1B}^2} = \frac{12.5}{100} \cdot \frac{24.15^2}{720} \cdot \frac{100}{23^2} = 0.019$$

$$Z_{T2}^* = \frac{U_{k2}\%}{100} \cdot \frac{U_{2N}^2}{S_N} \cdot \frac{S_B}{U_{2B}^2} = \frac{2.5}{100} \cdot \frac{24.15^2}{720} \cdot \frac{100}{23^2} = 3.94 \times 10^{-3}$$

$$Z_{T3}^* = \frac{U_{k3}\%}{100} \cdot \frac{U_{3N}^2}{S_N} \cdot \frac{S_B}{U_{3B}^2} = \frac{7.5}{100} \cdot \frac{24.15^2}{720} \cdot \frac{100}{23^2} = 0.012$$

$$\begin{cases} Z_{II1}^* = k_{13}^* \cdot Z_{T1}^* = 1.074 \times 0.019 = 0.02 \\ Y_{II130}^* = \frac{1 - k_{13}^*}{Z_{T1}^* \cdot k_{13}^{*2}} = \frac{1 - 1.074}{0.019 \times 1.074^2} = -3.38 \\ Y_{II310}^* = \frac{k_{13}^* - 1}{Z_{T1}^* \cdot k_{13}^*} = \frac{1.074 - 1}{0.019 \times 1.074} = 3.63 \end{cases}$$

$$\begin{cases} Z_{II2}^* = k_{23}^* \cdot Z_{T2}^* = 1.021 \times 3.94 \times 10^{-3} = 4.02 \times 10^{-3} \\ Y_{II230}^* = \frac{1 - k_{23}^*}{Z_{T2}^* \cdot k_{23}^{*2}} = \frac{1 - 1.021}{3.94 \times 10^{-3} \times 1.021^2} = -5.11 \\ Y_{II320}^* = \frac{k_{23}^* - 1}{Z_{T2}^* \cdot k_{23}^*} = \frac{1.021 - 1}{3.94 \times 10^{-3} \times 1.021} = 5.22 \end{cases}$$

线路：

$$X_{L1} = 0.4 \times 200 = 80\Omega, \quad X_{L1}^* = X_{L1} \cdot \frac{S_B}{U_{1B}^2} = 80 \cdot \frac{100}{500^2} = 0.032$$

$$X_{L2} = 0.4 \times 100 = 40\Omega, \quad X_{L2}^* = X_{L2} \cdot \frac{S_B}{U_{2B}^2} = 40 \cdot \frac{100}{220^2} = 0.083$$

第3章 作业

3-1

已知始端电压与功率： $\dot{U}_1 = 225\text{kV}$ ， $\tilde{S}_1 = 100 + j20\text{MVA}$

$$R = 0.131 \times 150 = 19.65\Omega, \quad X = 0.394 \times 150 = 59.1\Omega$$

(1) 电压降落计算：

$$d\dot{U} = \frac{P_1 R + Q_1 X}{U_1} + j \frac{P_1 X - Q_1 R}{U_1} = \frac{100 \times 19.65 + 20 \times 59.1}{225} + j \frac{100 \times 59.1 - 20 \times 19.65}{225}$$
$$d\dot{U} = 13.99 + j24.52$$

设始端电压相位为 0° ，则末端电压：

$$\dot{U}_2 = 225 - 13.99 - j24.52 = 211.01 - j24.52 = 212.43 \angle -6.62^\circ \text{kV}$$

幅值：212.43kV，相角： -6.62°

(2) 忽略横向分量：

$$\dot{U}_2 = 225 - 13.99 = 211.01\text{kV}$$

(3) 末端负荷功率损耗：

$$\Delta \tilde{S}_2 = \left(\frac{S_1^*}{U_1}\right)^2 (R + jX) = \frac{100^2 + 20^2}{225^2} (19.65 + j59.1) = 4.037 + j12.14\text{MVA}$$

末端功率：

$$\tilde{S}_2 = \tilde{S}_1 - \Delta \tilde{S}_2 = 95.96 + j7.86\text{MVA}$$

(4) 无功由 20 → 40MVA 电压降落：

$$d\dot{U}_1 = \frac{100 \times 19.65 + 40 \times 59.1}{225} + j \frac{100 \times 59.1 - 40 \times 19.65}{225} = 19.24 + j22.77\text{kV}$$

末端电压：

$$\dot{U}_2 = \dot{U}_1 - d\dot{U}_1 = 205.76 - j22.77 = 207.02 \angle -6.31^\circ \text{ kV}$$

功率损耗：

$$\Delta \tilde{S}_2 = \frac{100^2 + 40^2}{225^2} (19.65 + j59.1) = 4.5 + j13.54 \text{ MVA}$$

末端功率：

$$\tilde{S}_2 = \tilde{S}_1 - \Delta \tilde{S}_2 = 95.5 + j26.46 \text{ MVA}$$

3-2

$$\frac{Y}{2} = \frac{1}{2} \times 2.89 \times 150 = j216.75 \times 10^{-6} \text{ S}$$

首端并联导纳的功率损耗：

$$\Delta \tilde{S}_1 = \dot{U}_1 \cdot (\dot{U}_1 \cdot \frac{Y}{2})^* = U_1^2 \cdot (\frac{Y}{2})^* = (225 \times 10^3)^2 \cdot (-j216.75 \times 10^{-6}) = -j10.97 \text{ MVA}$$

末端并联导纳的功率损耗：

$$\Delta \tilde{S}_2 = \dot{U}_2 \cdot (\dot{U}_2 \cdot \frac{Y}{2})^* = U_2^2 \cdot (\frac{Y}{2})^* = (212.43 \times 10^3)^2 \cdot (-j216.75 \times 10^{-6}) = -j10.02 \text{ MVA}$$

$$\text{电压差： } \dot{U}_1 - \dot{U}_2 = 11.18 + j22.47 \text{ kV}$$

串联阻抗的功率损耗：

$$\Delta \tilde{S}_Z = \frac{|\dot{U}_1 - \dot{U}_2|^2}{R^2 + X^2} (R + jX) = \frac{11.18^2 + 22.47^2}{19.65^2 + 59.1^2} (19.65 + j59.1) = 3.191 + j9.592 \text{ MVA}$$

总功率损耗：

$$\Delta S = 3.191 + j9.592 - j10.97 - j10.02 = 3.191 - j11.398 \text{ MVA}$$

3-3

$$R = 0.131 \times 150 = 19.65 \Omega, \quad X = 0.394 \times 150 = 59.1 \Omega$$

$$\frac{Y}{2} = 2.89 \times 150 \times \frac{1}{2} = j216.75 \times 10^{-6} \text{ S}$$

$$\Delta \tilde{S}_1 = \dot{U}_1 \cdot \left(\dot{U}_1 \cdot \frac{Y}{2} \right)^* = U_1^2 \cdot \left(\frac{Y}{2} \right)^* = U_1^2 \left(-\frac{jB}{2} \right)$$

串联线路末端功率： $\tilde{S}'_2 = P'_2 + jQ'_2 = U_2^2 \left(-\frac{Y}{2} \right)$, ($Q'_2 = -\frac{1}{2}BU_2^2$, $P'_2 = 0$)

电压关系：

$$\dot{U}_1 = \dot{U}_2 + \frac{P'_2 R + Q'_2 X}{U_2} + j \frac{P'_2 X - Q'_2 R}{U_2} = \dot{U}_2 - \frac{U_2}{2} BX + j \frac{U_2}{2} BR$$

代入：

$$\begin{aligned} \dot{U}_1 &= \dot{U}_2 - U_2 \cdot 216.75 \times 10^{-6} \times 59.1 + j U_2 \cdot 216.75 \times 10^{-6} \times 19.65 \\ &= \dot{U}_2 - 0.0128 U_2 + j 0.00425 U_2 = (0.9872 + j 0.004259) U_2 \end{aligned}$$

解得末端电压：

$$\dot{U}_2 = \frac{\dot{U}_1}{0.9872 + j 0.004259} = 227.916 - j 0.983 \text{ kV} = 227.92 \angle -0.247^\circ \text{ kV}$$

3-4

极限输送公式：

$$P = \frac{U_1 U_2 \sin \theta}{X}, \quad \theta = \frac{\pi}{2} \text{ 时, } P_{\max} = \frac{U_1 U_2}{X}, \quad \text{即不计电阻 } R$$

$$220 \text{ kV 时: } P_{\max} = \frac{220^2}{80} = 605 \text{ MW}, \quad 500 \text{ kV 时: } P'_{\max} = \frac{500^2}{90} = 2777.78 \text{ MW}$$

标么值 ($S_B = 100 \text{ MVA}$):

$$P_{\max}^* = \frac{605}{100} = 6.05, \quad P'_{\max}^* = \frac{2777.78}{100} = 27.78$$

3-5

负荷功率因数为 1, $\tilde{S}_2 = 600 \text{ kW} = P_2$, 这里为方便计算, 忽略横向分量。

$$\text{电压损耗 } \Delta U_1 = \frac{U_1 - U_2}{U_N} = 10\% \implies \Delta U_1 = 0.1 U_N, \quad \text{此时 } U_2 = 3 \text{ kV}$$

$$\text{负荷功率因数为 } 0.8, \quad \tilde{S}' = P'_2 + j 0.75 P'_2 \text{ kVA}, \quad \Delta U_2 \leq 10\% \implies \Delta U_2 \leq 0.1 U_N$$

电压损耗：

$$\Delta U_1 = \frac{P_2 R + Q_2 X}{U_2} = \frac{600 R}{3} = 0.2 R \text{ kV} = 0.1 U_N$$

$$\Delta U_2 = \frac{P'_2 R + 0.75 P'_2 X}{U_2} = \frac{P'_2 R + 0.75 P'_2 \cdot 3R}{3} = \frac{1.75 P'_2 R}{3} (X = R)$$

由 $\frac{1.75 P'_2 R}{3} \leq 0.1 U_N = 0.2 R$, 得:

$$P' \leq 0.342 \text{ MW}$$

3-6

折算至高压侧: $U'_3 = U_3 \cdot \frac{110}{6.6} = 100 \text{ kV}$

阻抗: $z_{T1} = 0.7 + j65 \Omega$, $z_{T2} = 1.47 - j1.51 \Omega$, $z_{T3} = 2.47 + j37.8 \Omega$ (忽略横向分量)

1. ④ 电压:

$$U_4 = U'_3 + \frac{P_3 R_{T3} + Q_3 X_{T3}}{U'_3} = 100 + \frac{5 \times 2.47 + 3 \times 37.8}{100} = 101.2575 \text{ kV}$$

2. z_{T3} 损耗:

$$\Delta S_{T3} = \left(\frac{S_3^*}{U_3^*} \right)^2 z_{T3} = \frac{5^2 + 3^2}{100^2} (2.47 + j3.8) = 0.008398 + j0.12852 \text{ MVA}$$

$$3. \tilde{S}_{43} = \tilde{S}_3 + \Delta \tilde{S}_{T3} = 5.008398 + j3.12852 \text{ MVA}$$

4. 中压母线折算后电压设为 U'_2 , 则

$$U_4 = U'_2 + \frac{P_2 R_{T2} + Q_2 X_{T2}}{U'_2} = U'_2 + \frac{5 \times 1.47 - 1.51 \times 4}{U'_2} = U'_2 + \frac{1.31}{U'_2}$$

$$\therefore 101.2575 = U'_2 + \frac{1.31}{U'_2} \Rightarrow U'_2 = 101.245 \text{ kV}.$$

5. z_{T2} 损耗:

$$\Delta \tilde{S}_{T2} = \left(\frac{S_2^*}{U_2^*} \right)^2 z_{T2} = \frac{5^2 + 4^2}{101.245^2} (1.47 - j1.51) = 5.88 \times 10^{-3} - j6.04 \times 10^{-3} \text{ MVA}$$

$$6. \tilde{S}_{42} = \tilde{S}_2 + \Delta \tilde{S}_{T2} = 5.00588 + j3.99396 \text{ MVA}$$

$$7. \tilde{S}_{14} = \tilde{S}_{42} + \tilde{S}_{43} = 10.01428 + j7.12248 \text{ MVA}$$

$$8. U_1 = U_4 + \frac{P_{14} R_{T1} + Q_{14} X_{T1}}{U_4} = 101.2575 + \frac{10.01428 \times 0.7 + 7.12248 \times 65}{101.2575} = 105.899 \text{ kV}$$

$$9. \Delta \tilde{S}_{T1} = \left(\frac{S_{14}}{U_4^*} \right)^2 z_{T1} = \frac{10.01428^2 + 7.12248^2}{101.2575^2} (0.7 + j65) = 0.0103 + j0.95732$$

$$\tilde{S}_1 = \tilde{S}_{14} + \Delta \tilde{S}_{T1} = 10.01428 + 0.0103 + j7.12248 + j0.95732 = 10.02458 + j8.0798 \text{ MVA}.$$

$$\text{实际电压: } \begin{cases} U_1 = 105.899 \text{ kV.} \\ U_2 = U'_2 \left(\frac{38(1+0.05)}{110} \right) = 101.245 \times \frac{38 \times 1.05}{110} = 36.72 \text{ kV} \\ U_3 = 6 \text{ kV.} \end{cases}$$

3-7

根节点也叫平衡节点。根节点运行在 37 kV，计横向分量，不计 R 。

1. 计算 ① 电压 U_D :

$$U_D = U_1 - \left(\frac{P'_{L1} R_{L1} + Q'_{L1} X_{L1}}{U_1} + j \frac{P'_{L1} X_{L1} - Q'_{L1} R_{L1}}{U_1} \right) = U_1 - \frac{X_{L1}}{U_1} (Q'_{L1} + jP'_{L1})$$

$$\tilde{S}'_{L1} = \sum_{i=2}^5 \tilde{S}_i = 7.97 + j6.184 \text{ MVA, 则}$$

$$U_D = 37 - \frac{3.083}{37} (6.184 + j7.97) = 36.48 - j0.66 = 36.491 \angle -1.0^\circ$$

2. 计算 ② 电压 U_2 : $\tilde{S}_{L2} = \tilde{S}_2 + \tilde{S}_4 = 0.74 + j0.744 \text{ MVA}$

$$U_2 = U_D - \frac{X_{L3}}{U_D} (Q_{L2} + jP_{L2}) = 36.48 - j0.66 - \frac{1.305}{36.48 - j0.66} (0.744 + j0.74) = 36.45 - j0.687 \text{ kV}$$

3. 计算 ④ 的电压 U_4 :

$$U_4 = 36.45 - j0.687 - \frac{0.87}{36.45 - j0.687} (0.714 + j0.7) = 36.433 - j0.704 \text{ kV} = 36.44 \angle -1.07^\circ$$

4. 计算 ③ 的电压 U_3 : $\tilde{S}_{L3} = \tilde{S}_3 + \tilde{S}_5 = 0.83 + j0.64 \text{ MVA}$

$$U_3 = U_D - \frac{X_{L5}}{U_D} \tilde{S}_{L3}^* = 36.48 - j0.66 - \frac{1.74}{36.48 - j0.66} (0.64 + j0.83) = 36.45 - j0.68 \text{ kV.}$$

5. 计算 ⑤ 的电压 U_5 :

$$U_5 = U_3 - \frac{X_{L6}}{U_3} \tilde{S}_5^* = 36.45 - j0.68 - \frac{0.435}{36.45 - j0.68} (0.04 + j0.05) = 36.45 - j0.68 \text{ kV} = 36.456 \angle -1.06^\circ$$

故最大电压损耗发生在 ④ 点，则损耗 = $\frac{37 - 36.44}{35} \times 100\% = 1.60\%$

3-8

双端供电网络：自然功率分布 + 强制功率分布，近似计算时：

① 并联支路：假设节点电压为额定电压，算出运算负荷并入负荷。

② 串联支路：所有点均用额定电压。

1.B 打开时，看作两个独立的供电网络。左边 $\dot{U}_1 = \dot{U}_2$ ，右边 $\dot{U}_3 = \dot{U}_4 = \dot{U}_5$ 。

$$\Delta \tilde{S}_{Z12} = \left(\frac{S_2^*}{U_2^*}\right)^2 Z_{12} = \frac{P_2^2 + Q_2^2}{U_2^2} Z_{12} = \frac{30^2 + 14^2}{110^2} (12 + j30) = 1.087 + j2.717 \text{ MVA}$$

$$\tilde{S}_{12} = \tilde{S}_2 + \Delta \tilde{S}_{Z12} = 31.087 + j16.717 \text{ MVA}$$

$$\Delta \tilde{S}_{Z43} = \left(\frac{S_3^*}{U_3^*}\right)^2 Z_{34} = \frac{20^2 + 9^2}{120^2} (14 + j40) = 0.468 + j1.336 \text{ MVA}$$

$$\tilde{S}_{43} = \tilde{S}_3 + \Delta \tilde{S}_{Z43} = 20.468 + j10.336 \text{ MVA}$$

$$\tilde{S}'_4 = \tilde{S}_4 + \tilde{S}_{43} = 35.468 + j18.336 \text{ MVA}$$

$$\Rightarrow \Delta \tilde{S}_{Z54} = \left(\frac{\tilde{S}'_4}{U_4^*}\right)^2 Z_{45} = \frac{35.468^2 + 18.336^2}{120^2} (12 + j30) = 1.328 + j3.32 \text{ MVA}$$

$$\tilde{S}_{54} = \tilde{S}'_4 + \Delta \tilde{S}_{Z54} = 36.796 + j21.656 \text{ MVA}$$

2.B 闭合时，在 110 kV 电压级下：设 $U_B = 110 \text{ kV}$, $S_B = 100 \text{ MVA}$

$$U_1^* = 0.996 + j0.08, U_2^* = 1.074 + j0.189, U_1 = 109.56 + j8.8, U_2 = 118.14 + j20.79$$

自然功率分布：以 5 为支点，可求出 $\tilde{S}_{12}$ (自)

$$\begin{aligned} \tilde{S}_{12}^*(\text{自}) &= \frac{\tilde{S}_2^*(Z_{23} + Z_{34} + Z_{45}) + \tilde{S}_3^*(Z_{34} + Z_{45}) + \tilde{S}_4^* Z_{45}}{Z_{12} + Z_{23} + Z_{34} + Z_{45}} \\ &= 35.97 - j16.79 \text{ MVA} \end{aligned}$$

强制功率分布：

$$\begin{aligned} \tilde{S}_{12}^*(\text{强}) &= \frac{\dot{U}_1 - \dot{U}_5}{Z_{12} + Z_{23} + Z_{34} + Z_{45}} \times U_N \\ &= \frac{109.56 + j8.8 - 118.14 - j20.79}{58 + j159} \times 110 \\ &= -8.78 + j2.74 \text{ MVA} \\ \therefore \tilde{S}_{12} &= 27.19 + j14.05 \text{ MVA} \end{aligned}$$

同理：以 1 为支点，可求出 $\tilde{S}_{54}(\text{自})$

$$\tilde{S}_{54}^*(\text{自}) = \frac{\tilde{S}_4^*(Z_{12} + Z_{23} + Z_{34}) + \tilde{S}_3^*(Z_{12} + Z_{23}) + \tilde{S}_2^*Z_{12}}{Z_{12} + Z_{23} + Z_{34} + Z_{45}}$$

$$= 29.026 - j14.21 \text{ MVA}$$

$$\tilde{S}_{54}^*(\text{强}) = \frac{\dot{U}_5 - \dot{U}_1}{Z_{12} + Z_{23} + Z_{34} + Z_{45}} \times U_N$$

$$= 8.78 - j2.74 \text{ MVA}$$

$$\therefore \tilde{S}_{54} = 37.856 + j16.95 \text{ MVA}$$

功率分布：

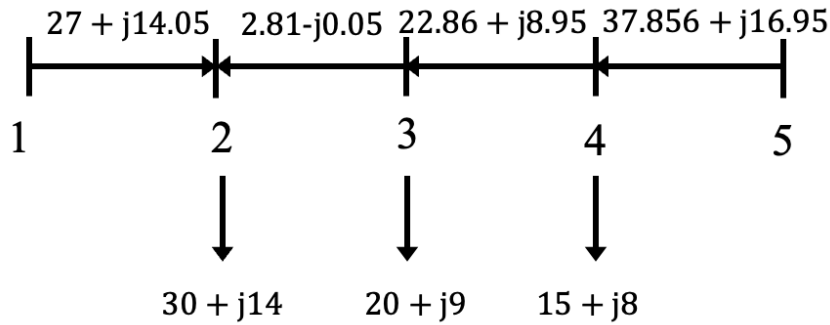


图 3.1: 功率流动示意图

有功功率的实际流动方向为 1 到 2 与 3 到 2，故 2 为有功功率分点。

无功功率的实际流动方向为 2 到 3 与 4 到 3，故 3 为无功功率分点。

3-9

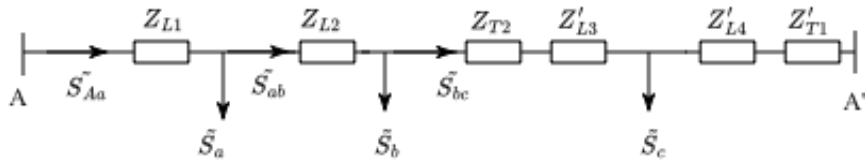


图 3.2: 拆环网

所有参数归算到同一电压等级 (209 kV)，取 $U_N = 220 \text{ V}$ ：

$$Z'_{L3} = \frac{209^2}{110^2} Z_{L3} \quad Z'_{L4} = \frac{209^2}{110^2} Z_{L4} \quad Z'_{T1} = \frac{209^2}{110^2} Z_{T1}$$

$$U'_A = 231 \cdot \frac{110}{231} \cdot \frac{209}{110} = 209 \text{ kV} \Rightarrow \text{双端供电}$$

$$S_{Aa}^* = \frac{S_a^*(Z_{L2} + Z_{T3} + Z'_{L3} + Z'_{L4} + Z'_{T1}) + S_b^*(Z_{T3} + Z'_{L3} + Z'_{L4} + Z'_{T1}) + S_c^*(Z'_{L4} + Z'_{T1})}{\sum Z} + \frac{(U_A - U'_A)U_N}{\sum Z}$$

$$\therefore S_{Aa} = 178.1408 + j118.6918 \text{ MVA}$$

$$S_{ab} = S_{Aa} - S_a = 128.1408 + j88.6918 \text{ MVA}$$

$$S_{bc} = S_{ab} - S_b = -21.8592 - j11.3082 \text{ MVA}$$



图 3.3: 打开功率分点

功率分点为 b , 左边:

$$S'_{ab} = S_{ab} + \Delta S_{L2} = 128.1408 + j88.6918 + \frac{128.1408^2 + 88.6918^2}{220^2} Z_{L2}$$

$$= 131.1515 + j113.781$$

$$S'_{Aa} = S'_{ab} + S_a + \Delta S_{L1} = 181.1515 + j143.781 + \frac{181.1515^2 + 143.781^2}{220^2} Z_{L1}$$

$$= 185.5721 + j187.9867$$

$$U_a = 231 - \left(\frac{185.5721 \times 4 + 187.9867 \times 40}{231} + j \frac{185.5721 \times 40 - 187.9867 \times 4}{231} \right)$$

$$= 185.23 - j28.8 = 197.36 \angle -8.41^\circ$$

$$U_b = 185.23 - j28.8 - \left[\frac{131.1515 \times 6 + 113.781 \times 50}{185.23 + j28.8} + j \frac{131.1515 \times 50 - 113.781 \times 6}{185.23 + j28.8} \right]$$

$$= 167.22 \angle -18.67^\circ$$

右边:

$$\begin{aligned}S'_{cb} &= S_{cb} + \Delta S_{T3} + \Delta S'_{L3} \\&= 21.8592 + j11.3082 + \frac{21.8592^2 + 11.3082^2}{220^2} (Z_{T3} + Z'_{L3}) \\&= 21.917 + j12.468\end{aligned}$$

$$\begin{aligned}S'_{AC} &= S'_{cb} + S_c + \Delta S_{T1} + \Delta S'_{L4} \\&= 21.917 + j22.468 + \frac{21.917^2 + 22.468^2}{220^2} (Z'_{L4} + Z_{T1}) \\&= 52.23 + j34.91\end{aligned}$$

$$\begin{aligned}U'_c &= 209 - \left[\frac{52.23 \times 5.5955 + 34.91 \times 205.77}{209} + j \frac{52.23 \times 205.77 - 34.91 \times 5.5955}{209} \right] \\&= 173.23 - j50.48 = 180.44 \angle -16.25^\circ\end{aligned}$$

$$U_c = \frac{110}{209} U'_c = 94.96 \angle -16.25^\circ$$

$$\begin{aligned}U_b &= 173.23 - j50.48 - \left(\frac{21.917 \times 4.61 + 12.468 \times 92.6875}{173.23 + j50.48} + j \frac{21.917 \times 92.6875 - 12.468 \times 4.61}{173.23 + j50.48} \right) \\&= 173.8187 \angle -19.85^\circ\end{aligned}$$

第 4 章 作业

注：第四章和第五章在 23 级的作业多了很多题，时间仓促，本章只写 22 级涉及部分。后续会逐渐完善，敬请理解！

4-1 (a)

节点导纳矩阵 $\mathbf{Y}$ ：

$$\begin{aligned}y_{10} &= Y_1 + \frac{1}{Z_1}, & y_{12} &= -\frac{1}{Z_1}, & y_{13} &= 0 \\y_{20} &= \frac{1}{Z_1} + \frac{1}{Z_2} + Y_1, & y_{21} &= -\frac{1}{Z_1}, & y_{23} &= -\frac{1}{Z_2} \\y_{30} &= \frac{1}{Z_2}, & y_{32} &= -\frac{1}{Z_2}, & y_{31} &= 0\end{aligned}$$

即：

$$\mathbf{Y} = \begin{bmatrix} Y_1 + \frac{1}{Z_1} & -\frac{1}{Z_1} & 0 \\ -\frac{1}{Z_1} & \frac{1}{Z_1} + \frac{1}{Z_2} + Y_1 & -\frac{1}{Z_2} \\ 0 & -\frac{1}{Z_2} & \frac{1}{Z_2} \end{bmatrix}$$

求 $\mathbf{Z}$ ：一般不用求逆矩阵，一般用单位电流注入法

① 注入 $I_1 = 1$ ， $I_2 = I_3 = 0$ ，则：

$$\begin{aligned}U_1 &= 1 \cdot \frac{1}{Y_1} \parallel (Z_1 + \frac{1}{Y_1}) = Z_{11} = \frac{1 + Y_1 Z_1}{Y_1(Z_1 Y_1 + 2)} \\Z_{12} = U_2 &= U_1 \frac{\frac{1}{Y_1}}{Z_1 + \frac{1}{Y_1}} = \frac{1 + Y_1 Z_1}{Y_1(Z_1 Y_1 + 2)} \cdot \frac{1}{Z_1 Y_1 + 1} = \frac{1}{Y_1(Z_1 Y_1 + 2)} \\Z_{13} = U_3 &= U_2 = \frac{1}{Y_1(Z_1 Y_1 + 2)}\end{aligned}$$

② 注入 $I_2 = 1$ ， $I_1 = I_3 = 0$ ，则：

$$Z_{22} = U_2 = 1 \cdot \frac{1}{Y_1} \parallel (Z_1 + \frac{1}{Y_1}) = \frac{1 + Y_1 Z_1}{Y_1(Z_1 Y_1 + 2)}$$

$$Z_{21} = U_2 \cdot \frac{\frac{1}{Y_1}}{Z_1 + \frac{1}{Y_1}} = \frac{1}{Y_1(Z_1 Y_1 + 2)}$$

$$Z_{23} = Z_{22} = \frac{1 + Y_1 Z_1}{Y_1(Z_1 Y_1 + 2)}$$

③ 注入 $I_3 = 1$, $I_2 = I_1 = 0$, 则:

$$Z_{33} = U_3 = Z_2 + \frac{1}{Y_1} \parallel (Z_1 + \frac{1}{Y_1}) = Z_2 + \frac{1 + Y_1 Z_1}{Y_1(Z_1 Y_1 + 2)}$$

$$Z_{32} = U_2 = \frac{1 + Y_1 Z_1}{Y_1(Z_1 Y_1 + 2)}$$

$$Z_{31} = U_1 = \frac{1}{Y_1(Z_1 Y_1 + 2)}$$

最终结果:

$$\Rightarrow Z = \begin{bmatrix} \frac{1 + Y_1 Z_1}{Y_1(Z_1 Y_1 + 2)} & \frac{1}{Y_1(Z_1 Y_1 + 2)} & \frac{1}{Y_1(Z_1 Y_1 + 2)} \\ \frac{1}{Y_1(Z_1 Y_1 + 2)} & \frac{1 + Y_1 Z_1}{Y_1(Z_1 Y_1 + 2)} & \frac{1 + Y_1 Z_1}{Y_1(Z_1 Y_1 + 2)} \\ \frac{1}{Y_1(Z_1 Y_1 + 2)} & \frac{1 + Y_1 Z_1}{Y_1(Z_1 Y_1 + 2)} & Z_2 + \frac{1 + Y_1 Z_1}{Y_1(Z_1 Y_1 + 2)} \end{bmatrix}$$

(b)

节点导纳矩阵 $\mathbf{Y}$:

$$\begin{aligned} Y_{11} &= \frac{1}{Z_1} + \frac{1}{Z_3}, & Y_{12} &= -\frac{1}{Z_1}, & Y_{13} &= -\frac{1}{Z_3} \\ Y_{21} &= -\frac{1}{Z_1}, & Y_{22} &= \frac{1}{Z_1} + \frac{1}{Z_2}, & Y_{23} &= -\frac{1}{Z_2} \\ Y_{31} &= -\frac{1}{Z_3}, & Y_{32} &= -\frac{1}{Z_2}, & Y_{33} &= \frac{1}{Z_2} + \frac{1}{Z_3} \end{aligned}$$

即:

$$\mathbf{Y} = \begin{bmatrix} \frac{1}{Z_1} + \frac{1}{Z_3} & -\frac{1}{Z_1} & -\frac{1}{Z_3} \\ -\frac{1}{Z_1} & \frac{1}{Z_1} + \frac{1}{Z_2} & -\frac{1}{Z_2} \\ -\frac{1}{Z_3} & -\frac{1}{Z_2} & \frac{1}{Z_2} + \frac{1}{Z_3} \end{bmatrix}$$

注意: 无接地支路, 则 $\mathbf{Y}$ 不满秩, $\mathbf{Z}$ 不存在 (或元素均为无穷大)。

探讨: 为何无接地支路时 $\mathbf{Y}$ 不满秩?

假设无接地支路时满秩，则 $Y\dot{U} = \dot{I}$ ， $\dot{U} = Y^{-1}\dot{I}$ 。针对给定的电流向量 $\dot{I}$ 可以求出唯一对应的电压向量 $\dot{U}$ 。对于 $\dot{U} + cI_n$ (I_n 为全 1 向量，相当于此时 U 整体浮动)：

$$I_n = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, \quad u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \rightarrow \begin{bmatrix} u_1 + c \\ u_2 + c \\ \vdots \\ u_n + c \end{bmatrix}, \text{ 证明 } Y(\dot{U} + cI_n) \text{ 同样 } = \dot{I} \text{ 即可。}$$

$$\text{即 } YcI_n = 0. \text{ 由节点导纳阵: } YI_n = \begin{bmatrix} Y_{11} & Y_{12} & \cdots & Y_{1n} \\ Y_{21} & \cdots & & \vdots \\ Y_{31} & \cdots & & \\ \vdots & & & \\ Y_{n1} & \cdots & \cdots & Y_{nn} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = Y_{11} + Y_{12} + \cdots + Y_{1n}.$$

设 1 与支路 $k_1, k_2 \cdots k_m$ 相连，则 $Y_{11} = Y_{1k_1} + Y_{1k_2} + \cdots + Y_{1k_m}$ 。

而 $Y_{12} + Y_{13} + \cdots + Y_{1n}$ 为互导纳相反数之和，即为 $-Y_{1k_1} + (-Y_{1k_2}) + \cdots + (-Y_{1k_m})$ 。

$\therefore YI_n = 0$ ，即 $Y(\dot{U} + cI_n) = Y\dot{U} + cYI_n = Y\dot{U} + 0 = \dot{I}$ 。

即对于一个注入电流 $\dot{I}$ ，有多个解，故矛盾，不可逆。

倘若有接地支路，则 $Y_{11} = Y_{1k_1} + Y_{1k_2} + \cdots + Y_{1k_m} + Y_{10} \rightarrow$ 接地导纳。

$YI_n = 0 + Y_{10} = Y_{10}$ ，不为 0，故无法满足 $Y(\dot{U} + cI_n) = \dot{I}$ 。

4-6

(1) 形成节点导纳阵：

$$Y_{11} = \frac{1}{0.01 + j0.02} = 20 - j40, \quad Y_{12} = 0, \quad Y_{13} = -20 + j40.$$

$$Y_{21} = 0, \quad Y_{22} = 20 - j40, \quad Y_{23} = -20 + j40.$$

$$Y_{31} = -20 + j40, \quad Y_{32} = -20 + j40, \quad Y_{33} = 40 - j80.$$

$$\Rightarrow Y = \begin{bmatrix} 20 - j40 & 0 & -20 + j40 \\ 0 & 20 - j40 & -20 + j40 \\ -20 + j40 & -20 + j40 & 40 - j80 \end{bmatrix}$$

(2) 归纳未知数与方程个数:

PQ: 2 未知数, 2 方程, PV: 1 未知数, 1 方程

$$(3) \begin{bmatrix} \Delta P_2 \\ \Delta P_3 \\ \Delta Q_3 \end{bmatrix} = - \begin{bmatrix} H_{22} & H_{23} & N_{23} \\ H_{32} & H_{33} & N_{33} \\ M_{32} & M_{33} & L_{33} \end{bmatrix} \begin{bmatrix} \Delta \theta_2 \\ \Delta \theta_3 \\ \Delta u_3/u_3 \end{bmatrix} \quad \begin{array}{l} \text{设 } \dot{u}_2 = 1.02 \angle 0^\circ, \dot{u}_3 = 1 \angle 0^\circ \\ \text{开始套公式如下} \end{array}$$

$$\Delta P_2 = P_{G2} - P_{L2} - u_2 \left[\sum_{j=1}^n u_j (G_{2j} \cos \theta_{2j} + B_{2j} \sin \theta_{2j}) \right]$$

$$= 0.4 - 1.02[(1 \times 0 + 0 \times 0) + 1.02(20 \times 1 + 0) + 1 \times (-20)]$$

$$= 0.4 - 1.02(1.02 \times 20 - 20) = 0.4 - 1.02 \times 0.02 \times 10 = -0.008$$

$$\Delta P_3 = -1 - 1 \times [1 \times (-20) + 1.02 \times (-20) + 1 \times 40]$$

$$= -1 - (-20 - 20.4 + 40) = -1 - (-0.4) = -0.6$$

$$\Delta Q_3 = -0.5 - 1 \times [1 \times (-40) + 1.02 \times (-40) + 1 \times (-80)]$$

$$= -0.5 - [-40 - 40.8 + 80] = 0.3$$

$$H_{22} = \frac{\partial P_2}{\partial \theta_2} = u_2 u_1 G_{21} \sin \theta_{21} - u_2 u_1 B_{21} \cos \theta_{21} + u_2 u_3 G_{23} \sin \theta_{23} - u_2 u_3 B_{23} \cos \theta_{23}$$

$$= 1.02 \cdot 1.0 \cdot 0 - 0 + 1.02 \cdot 1 \cdot (-20) \cdot 0 - 1.02 \cdot 1 \cdot 40 \cdot 1 = -40.8$$

$$H_{23} = \frac{\partial P_2}{\partial \theta_3} = -u_2 u_3 G_{23} \sin \theta_{23} + u_2 u_3 B_{23} \cos \theta_{23}$$

$$= -1.02 \times 1 \times (-20) \times 0 + 1.02 \times 1 \times 40 = 40.8$$

$$H_{32} = \frac{\partial P_3}{\partial \theta_2} = -u_3 u_2 G_{32} \sin \theta_{32} + u_3 u_2 B_{32} \cos \theta_{32}$$

$$= -1 \times 1.02 \times (-20) \times 0 + 1 \times 1.02 \times 40 \times 1 = 40.8$$

$$H_{33} = \frac{\partial P_3}{\partial \theta_3} = u_3 u_1 G_{31} \sin \theta_{31} - u_3 u_1 B_{31} \cos \theta_{31} + u_3 u_2 G_{32} \sin \theta_{32} - u_3 u_2 B_{32} \cos \theta_{32}$$

$$= 1 \times 1 \times 0 - 1 \times 1 \times 40 \times 1 - 1 \times 1.02 \times 40 = -80.8$$

$$N_{23} = \frac{\partial P_2}{\partial u_3} u_3 = -u_2 u_3 (G_{23} \cos \theta_{23} + B_{23} \sin \theta_{23})$$

$$= -1.02 \times 1 \times (-20) = 20.4$$

$$N_{33} = \frac{\partial P_3}{\partial u_3} u_3 = -u_3 u_1 (G_{31} \cos \theta_{31} + B_{31} \sin \theta_{31}) - u_3 u_2 (G_{32} \cos \theta_{32} + B_{32} \sin \theta_{32}) - 2u_3^2 G_{33}$$

$$= -1 \times (-20) - 1 \times 1.02 \times (-20) - 2 \times 1 \times 40 = 40.4 - 80 = -39.6$$

$$M_{32} = \frac{\partial Q_3}{\partial \theta_2} = u_3 u_2 G_{32} \cos \theta_{32} + u_3 u_2 B_{32} \sin \theta_{32}$$

$$= 1 \times 1.02 \times (-20) \times 1 = -20.4$$

$$M_{33} = \frac{\partial Q_3}{\partial \theta_3} = -u_3 u_1 G_{31} \cos \theta_{31} - u_3 u_1 B_{31} \sin \theta_{31} - u_3 u_2 G_{32} \cos \theta_{32} - u_3 u_2 B_{32} \sin \theta_{32}$$

$$= -1 \times 1 \times (-20) \times 1 - 1 \times 1.02 \times (-20) \times 1 = 20 + 20.4 = 40.4$$

$$L_{33} = \frac{\partial Q_3}{\partial u_3} u_3 = -u_3 u_1 G_{31} \sin \theta_{31} + u_3 u_1 B_{31} \cos \theta_{31} - u_3 u_2 G_{32} \sin \theta_{32} + u_3 u_2 B_{32} \cos \theta_{32} + 2u_3^2 B_{33}$$

$$= 1 \times 1 \times 40 + 1 \times 1.02 \times 40 + 2 \times 1^2 \times (-80) = -79.2$$

$$\Rightarrow \begin{bmatrix} -0.008 \\ -0.6 \\ 0.3 \end{bmatrix} = - \begin{bmatrix} -40.8 & 40.8 & 20.4 \\ 40.8 & -80.8 & -39.6 \\ -20.4 & 40.4 & -79.2 \end{bmatrix} \begin{bmatrix} \Delta \theta_2 \\ \Delta \theta_3 \\ \Delta u_3/u_3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \Delta \theta_2 \\ \Delta \theta_3 \\ \Delta u_3/u_3 \end{bmatrix} = \begin{bmatrix} 0.002 \\ 0.015 \\ -0.0075 \end{bmatrix} \quad \text{故第一轮迭代结果: } \begin{cases} \dot{u}_2 = 1.02 \angle 0.002 \\ \dot{u}_3 = 0.9925 \angle 0.015 \end{cases}$$

4-6

(1) ①: PQ 节点, 2 个未知数, 2 个方程, ②: PV 节点, 1 个未知数, 1 个方程

③: 平衡节点, 无

(2) 节点导纳阵:

$$\begin{aligned}
 Y_{11} &= \frac{2}{j0.1} = -20j, & Y_{12} &= 10j, & Y_{13} &= 10j \\
 Y_{21} &= 10j, & Y_{22} &= -20j, & Y_{23} &= 10j \\
 Y_{31} &= 10j, & Y_{32} &= 10j, & Y_{33} &= -20j
 \end{aligned}$$

$$Y = \begin{bmatrix} -20j & 10j & 10j \\ 10j & -20j & 10j \\ 10j & 10j & -20j \end{bmatrix}, \quad \text{无 } G_{ij} \quad \text{设初始: } u_1 = 1\angle 0^\circ, \quad u_2 = 1\angle 0^\circ$$

(3) 套公式:

$$\begin{aligned}
 \Delta P_1 &= P_{G1} - P_{L1} - u_1 \sum_{j=1}^n u_j B_{1j} \sin \theta_{1j} \\
 &= -2 - 0 = -2 \\
 \Delta P_2 &= P_{G2} - P_{L2} - u_2 \sum_{j=1}^n u_j B_{2j} \sin \theta_{2j} \\
 &= 1 - 0 - 0 = 1 \\
 \Delta Q_1 &= Q_{G1} - Q_{L1} + u_1 \sum_{j=1}^n u_j B_{1j} \cos \theta_{1j} \\
 &= -1 + 1[1 \times (-20) + 10 + 10] = -1
 \end{aligned}$$

$$\therefore \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix} = - \begin{bmatrix} H_{11} & H_{12} & N_{11} \\ H_{21} & H_{22} & N_{21} \\ M_{11} & M_{12} & L_{11} \end{bmatrix} \begin{bmatrix} \Delta \theta_1 \\ \Delta \theta_2 \\ \Delta u_1 / u_1 \end{bmatrix}$$

$$H_{11} = \frac{\partial P_1}{\partial \theta_1} = -u_1 u_2 B_{12} \cos \theta_{12} - u_1 u_3 B_{13} \cos \theta_{13} = -10 - 10 = -20$$

$$H_{12} = \frac{\partial P_1}{\partial \theta_2} = u_1 u_2 B_{12} \cos \theta_{12} = 10$$

$$H_{21} = \frac{\partial P_2}{\partial \theta_1} = u_2 u_1 B_{21} \cos \theta_{21} = 10$$

$$H_{22} = \frac{\partial P_2}{\partial \theta_2} = -u_2 u_1 B_{21} \cos \theta_{21} - u_2 u_3 B_{23} \cos \theta_{23} = -10 - 10 = -20$$

$$\begin{aligned}
N_{11} &= \frac{\partial P_1}{\partial u_1} u_1 = -u_1 u_2 B_{12} \sin \theta_{12} - u_1 u_3 B_{13} \sin \theta_{13} = 0 \\
N_{21} &= \frac{\partial P_2}{\partial u_1} u_1 = -u_2 u_1 B_{21} \sin \theta_{21} = 0 \\
M_{11} &= \frac{\partial Q_1}{\partial \theta_1} = -u_1 u_2 B_{12} \sin \theta_{12} - u_1 u_3 B_{13} \sin \theta_{13} = 0 \\
M_{12} &= \frac{\partial Q_1}{\partial \theta_2} = u_1 u_2 B_{12} \sin \theta_{12} = 0 \\
L_{11} &= \frac{\partial Q_1}{\partial u_1} u_1 = 2u_1^2 B_{11} + u_1 u_2 B_{12} \cos \theta_{12} + u_1 u_3 B_{13} \cos \theta_{13}
\end{aligned}$$

$$= 2(-20) + 10 + 10 = -20$$

$$\therefore \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix} = - \begin{bmatrix} -20 & 10 & 0 \\ 10 & -20 & 0 \\ 0 & 0 & -20 \end{bmatrix} \begin{bmatrix} \Delta \theta_1 \\ \Delta \theta_2 \\ \Delta u_1 / u_1 \end{bmatrix}$$

即：

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} \Delta \theta_1 \\ \Delta \theta_2 \\ \Delta u_1 \end{bmatrix} = \begin{bmatrix} -0.2 \\ 0.1 \\ -0.1 \end{bmatrix}$$

$$\Rightarrow \begin{cases} \Delta \theta_1 = -0.1 \\ \Delta \theta_2 = 0 \\ \Delta u_1 = -0.05 \end{cases}, \text{第一次迭代结果} \begin{cases} \dot{u}_1 = 0.95 \angle -0.1 \\ \dot{u}_2 = 1 \angle 0^\circ \end{cases}$$

 **笔记** 大家最好在计算器上亲手操作一下如何求逆矩阵。

4-9

(1) 直接套公式：

$$H : \begin{cases} H_{ij} = u_i u_j B_{ij} \\ H_{ii} = u_i \sum_{j \in i, j \neq i} u_j (-B_{ij}) \end{cases} \quad N : \begin{cases} N_{ij} = 0 \\ N_{ii} = 0 \end{cases}$$

$$M : \begin{cases} M_{ij} = 0 \\ M_{ii} = 0 \end{cases} \quad L : \begin{cases} L_{ij} = H_{ij} = u_i u_j B_{ij} \\ L_{ii} = u_i \sum_{j \in i, j \neq i} u_j B_{ij} + 2u_i^2 B_{ii} \end{cases}$$

(2) 不等价： $G = 0$ 只是在计算时方便处理了矩阵。而 $R = 0$ 是完全忽略了线路电阻，让电压降、功率损耗的计算均会有影响，完全消除了热效应。

4-10

节点导纳阵:

$$y_{12} = \frac{1}{0.01 + j0.1} = 0.9901 - j9.901, \quad \frac{b_1}{2} = 0.0005$$

$$y_{23} = \frac{1}{0.02 + j0.1} = 1.92 - j9.62, \quad \frac{b_2}{2} = 0.001$$

$$y_{31} = \frac{1}{0.03 + j0.1} = 2.75 - j9.17, \quad \frac{b_3}{2} = 0.0002$$

$$\therefore Y = \begin{bmatrix} 3.74 - j19.07 & -0.9901 + j9.901 & -2.75 + j9.17 \\ -0.9901 + j9.901 & 2.91 - j19.52 & -1.92 + j9.62 \\ -2.75 + j9.17 & -1.92 + j9.62 & 4.67 - j18.79 \end{bmatrix}$$

$$B' = - \begin{bmatrix} -19.07 & 9.901 \\ 9.901 & -19.52 \end{bmatrix} = \begin{bmatrix} 19.07 & -9.901 \\ -9.901 & 19.52 \end{bmatrix}$$

$$B'' = - \begin{bmatrix} -19.07 \end{bmatrix} = \begin{bmatrix} 19.07 \end{bmatrix}$$

第5章 作业

5-1 (可以看学辅的电分资料 P42)

发电机不采取调频: $K_G = 0$ 。

$$\Delta f = \frac{\Delta P_L}{K_L} = 0.2 \text{ Hz}$$

$$f' = f_N - \Delta f = 49.8 \text{ Hz}$$

负荷功率特性: $P_L = 25000 \text{ MW} + 200 \text{ MW} \rightarrow 25200 \text{ MW}$

发电机功率特性: 但实际上由图, 仍为 25000 MW 。

5-2



笔记 当一台发电机达到满载后, 调节能力达到上限, 不再参与一次调频。因为调速器的基本原理是当频率 f 下降时, 增加进汽/进水量来提升出力。如果发电机已经满载, 阀门已开到最大, 即使频率继续下降, 它也无法再增加任何功率。

先由 $\sigma\% \rightarrow K_G$:

$$\begin{aligned} K_G &= \frac{100}{\sigma\%} \times \frac{P_{GN}}{f_N} \\ &= \frac{100}{4} \times \frac{100}{50} = 50 \text{ MW/Hz} \end{aligned}$$

(1) 3 台满载, $P_L \uparrow$ 时, 全由剩下一台机组承担。

1 : 100 MW, 2 : 100 MW, 3 : 100 MW, 4 : 20 MW $\rightarrow$ 100 MW。

$$\begin{aligned} \therefore \Delta f &= -\frac{\Delta P_L}{K_G} \\ &= -\frac{100 - 20}{50} = -\frac{80}{50} = -1.6 \text{ Hz} \\ f' &= f_N + \Delta f = 48.4 \text{ Hz} \end{aligned}$$

(2) 4 台平均承担时, 每台 $P_G = 320/4 = 80 \text{ MW}$ 。

相当于每台均由 80 MW $\rightarrow$ 100 MW。

$$\begin{aligned}\Delta f &= -\frac{\Delta P_L}{K_G} \\ &= -\frac{100 - 80}{50} = -0.4 \text{ Hz} \\ f' &= f + \Delta f = 49.6 \text{ Hz}\end{aligned}$$

(3) 同上，相当于每台均由 80 MW $\rightarrow$ 0。

$$\begin{aligned}\Delta f &= -\frac{\Delta P_L}{K_G} \\ &= -\frac{80}{50} = -1.6 \text{ Hz} \\ \therefore f' &= f + \Delta f = 48.4 \text{ Hz}\end{aligned}$$

5-3

此题需要注意满载情况！

$$\begin{aligned}K_{G1} &= \frac{100}{\sigma_1 \%} \cdot \frac{100}{50} = 100 \text{ MW/Hz} \\ K_{G2} &= \frac{100}{\sigma_2 \%} \cdot \frac{100}{50} = \frac{100}{3} = 33.333 \text{ MW/Hz} \\ K_{G3} &= \frac{100}{\sigma_3 \%} \cdot \frac{100}{50} = 40 \text{ MW/Hz} \rightarrow \text{已满载} \\ K_{G\Sigma} &= K_{G1} + K_{G2} = 133.333 \text{ MW/Hz}\end{aligned}$$

(1) 假设 1, 2 机组都未满载或刚好满载。

$$\begin{aligned}\Delta f_1 &= -\frac{\Delta P_L}{K_{G\Sigma} + K_L} \\ &= -\frac{50}{133.333 + 6.667} = -\frac{50}{140} = -0.357 \text{ Hz}\end{aligned}$$

验算：

$$\Delta P_{G1} = -K_{G1} \cdot \Delta f = 35.7 \text{ MW} \leq 100 - 60 = 40 \text{ MW}$$

$$\Delta P_{G2} = -K_{G2} \cdot \Delta f = 11.899 \text{ MW} < 100 - 80 = 20 \text{ MW}$$

$$\text{故 } f' = 50 - 0.357 = 49.643 \text{ Hz}$$

(2) 先判断哪台机组先满载：

$$1 \text{ 满载时, } \Delta f = -\frac{\Delta P_L}{K_{G1}} = -\frac{100 - 60}{100} = -0.4 \text{ Hz}$$

$$2 \text{ 满载时, } \Delta f_2 = -\frac{\Delta P_L}{K_{G2}} = -\frac{100 - 80}{33.333} = -0.6 \text{ Hz,}$$

$\therefore |\Delta f_1| = 0.4 \text{ Hz}$ 时, 1 已满载, 计算 2 此时还差多少满载:

$$P_{G2} = 80 + 0.4 \times 33.333 = 93.33 \text{ MW, 还差 } 6.67 \text{ MW.}$$

$$\Delta f' = -\frac{\Delta P'_{G2}}{K_{G2}} = -\frac{6.67}{33.333} \text{ Hz} = -0.2 \text{ Hz.}$$

$$\text{频率减少部分对应的 } \Delta P_L = K_L \Delta f' = 6.667 \times 0.6 = 4 \text{ MW.}$$

额定到达的负荷应为 $P_{LN} = 300 + 4 \text{ MW} = 304 \text{ MW}$, 实际只增加到了 300 MW .

5-4



笔记 15% 的热备用容量意味着该机组并没有满负荷运行, 而是预留了一部分调节的空间, 机组实际出力只有额定值的 85%。

额定负荷 P_{LN} , 最大机组: $P_{GN} = 0.1 P_{LN}$.

$$\begin{aligned} \therefore K_G &= \frac{100}{\sigma\%} \times \frac{P_{GN}}{f_N} = \frac{100}{5} \times \frac{0.1 P_{LN}}{50} \\ &= 0.04 P_{LN} \end{aligned}$$

由于最大机组留有热备用, 故实际出力 $P_G = 0.85 P_{GN} = 0.085 P_{LN}$

$$\Delta f = \frac{0.05 P_{LN}}{K_G + K'_L} = 0.1 \Rightarrow K'_L + K_G = 0.5 P_{LN}, \quad K'_L = 0.46 P_{LN}.$$

最大机组因故障退出时, 自然也没有了调节能力:

$$\begin{aligned} \Delta f' &= -\frac{0.085 P_{LN}}{0.46 P_{LN}} \\ &= -0.185 \text{ Hz} \end{aligned}$$

5-5

$$\text{套公式即可, } K_A = \frac{\Delta P_L}{\Delta f} = 2500 \text{ MW/Hz, } K_B = \frac{\Delta P_L}{\Delta f} = 4000 \text{ MW/Hz.}$$

对于 A: $P_{LA} + \Delta P_{AB} - P_{GA} = K_A \Delta f$, 联合运行后, 统一频率为 $f (49.85 < f < 50) \text{ Hz}$

对于 B: $P_{LB} - \Delta P_{AB} - P_{GB} = K_B \Delta f$

且由题: $\Delta P_A = \Delta P_B = 0$,

$$\begin{cases} -\Delta P_{AB} = 4000(50 - f) \\ \Delta P_{AB} = 2500(49.85 - f) \end{cases} \Rightarrow \text{解得 } f = 49.9423076 \text{ Hz}$$

$$\therefore \Delta P_{BA} = -\Delta P_{AB} = 2500(f - 49.85) = 230.769 \text{ MW}$$

5-6(此题很重要! 注意看题目给的提示)

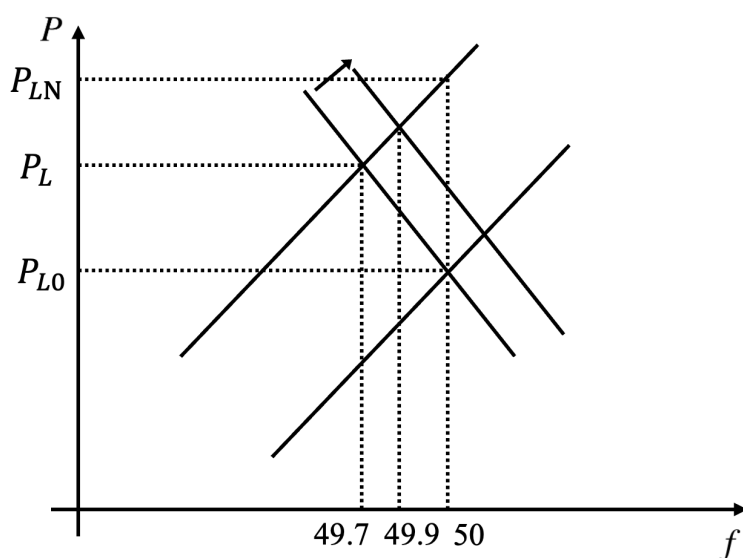


图 5.1: 二次调频示意图

(1)

$$P_{FN} = 0.2P_{L0}, \quad P_L = 1.1P_{L0}, \quad K_L = K_L^* \cdot \frac{P_{LN}}{f_N} = 2 \cdot \frac{1}{50} P_{LN} = \frac{P_{LN}}{25}$$

$$P_{LN} - P_L = K_L \Delta f \Rightarrow P_{LN} - 1.1P_{L0} = \frac{P_{LN}}{25} \times 0.3$$

$$\therefore P_{LN} = \frac{25 \times 1.1}{25 - 0.3} P_{L0} = 1.113P_{L0}.$$

系统回升 0.2Hz 后, 由 $P_{LN} - P_{L0}$ 段构成关系:

$$P_G'' + K_S \cdot 0.1 = P_{LN} - P_{L0} = 0.113P_{L0}$$

$$\text{而 } K_S = \frac{P_{LN} - P_{L0}}{\Delta f} = \frac{0.113P_{L0}}{0.3} \Rightarrow P_G'' = 0.07533P_{L0} = 0.377P_{FN}.$$

(2) 3 部分出力:

$$0.5P_{fN} + 0.378P_{fN} + 0.1K_G \leq P_{fN} \Rightarrow K_G \leq 1.22P_{fN}.$$

(原始出力+ 二次调频+ 一次调频)

$$\Rightarrow K_G \cdot \frac{f_N}{P_{FN}} \leq 1.22 \cdot P_{fN} \cdot \frac{f_N}{P_{FN}}, \text{ 即 } K_G^* \leq 1.22f_N = 1.22 \times 50 = 61 \text{ 即可}$$

$$\text{由 } \sigma\%_{\text{水}} = 2 - 4\% \quad \sigma\%_{\text{汽}} = 4 - 6\%$$

$$K_{G\text{水}}^* = 50 - 25 \quad K_{G\text{汽}}^* = 25 - 16, \text{ 满足条件, 故可以}$$

5-7

$$\begin{aligned} R_T &= \frac{P_k}{1000S_N} \cdot \frac{U_N^2}{S_N} \\ &= \frac{163 \times 110^2}{1000 \times 20 \times 20} = 4.93\Omega \\ X_T &= \frac{U_k\%}{100} \cdot \frac{U_N^2}{S_N} \\ &= \frac{10.5}{100} \cdot \frac{110^2}{20} = 63.525\Omega. \end{aligned}$$

$$\text{最大负荷时, } P_{\max} + jQ_{\max} = 14.4 + j10.8 \text{ MVA}$$

$$\text{最小负荷时, } P_{\min} + jQ_{\min} = 4.9 + j5 \text{ MVA}$$

$$U'_{2\max} = 107.5 - \frac{14.4 \times 4.93 + 10.8 \times 63.525}{107.5} - j \frac{14.4 \times 63.525 - 10.8 \times 4.93}{107.5}$$

$$\text{忽略横分量, } U_{2\max} = 107.5 - 7.04 = 100.46 \text{ kV}$$

$$U'_{2\min} = 107.5 - \frac{4.9 \times 4.93 + 5 \times 63.525}{107.5} = 104.32 \text{ kV}$$

低压侧母线偏移: $1.025 \sim 1.075U_{2N}$, 则为顺调压

要求: 最大负荷时, $U_2 \geq 1.025U_{2N}$, 最小负荷时, $U_2 \leq 1.075U_{2N}$ 。

$$\text{最大负荷时, } \frac{U_{2\max}}{1.025U_{2N}} = \frac{U_{t\max}}{11} \Rightarrow U_{t\max} = \frac{100.46}{1.025 \times 10} \times 11 = 107.81 \text{ kV}$$

$$\text{最小负荷时, } \frac{U_{2\min}}{1.075U_{2N}} = \frac{U_{t\min}}{11} \Rightarrow U_{t\min} = \frac{104.32}{1.075 \times 10} \times 11 = 106.75 \text{ kV} < U_{t\max}$$

$$U_t = \frac{1}{2}(U_{t\max} + U_{t\min}) = \frac{1}{2} \times (107.81 + 106.75) = 107.28 \text{ kV}$$

选择：107.25 kV 分接头，再用这个电压回代，看低压侧母线电压是否在范围内。

$$U_{2\min} = \frac{11 \times 104.32}{107.25} = 10.699 \text{ kV}$$

$$U_{2\max} = \frac{11 \times 100.46}{107.25} = 10.30 \text{ kV}$$

两个电压均位于 (10.25, 10.75) kV 内。

5-8

升压变压器计算高压侧 U'_1 电压：

$$\text{最大负荷: } U'_{1\max} = U_{1\max} + \frac{2.1 \times 28 + 38.5 \times 21}{U_{1\max}} = 112.09 + \frac{2.1 \times 28 + 38.5 \times 21}{112.09} = 119.83 \text{ kV}$$

$$\text{最小负荷: } U'_{1\min} = U_{1\min} + \frac{2.1 \times 14 + 38.5 \times 10.5}{U_{1\min}} = 115.45 + \frac{2.1 \times 14 + 38.5 \times 10.5}{115.45} = 119.21 \text{ kV}$$

要求：低压侧母线电压为 10 kV ~ 11 kV，恒调压。

$$\text{最大负荷时, } \frac{10.5}{U_{t\max}} = \frac{10}{119.83} \Rightarrow U_{t\max} = 125.82 \text{ kV}$$

$$\text{最小负荷时, } \frac{10.5}{U_{t\min}} = \frac{11}{119.21} \Rightarrow U_{t\min} = 113.79 \text{ kV}$$

$$\Rightarrow U_t = \frac{1}{2}(U_{t\max} + U_{t\min}) = 119.36 \text{ kV, 选121 kV 主接头, 回代验算:}$$

$$U_{2\max} = \frac{10.5 \times 119.83}{121} = 10.39 \text{ kV}$$

$$U_{2\min} = \frac{10.5 \times 119.21}{121} = 10.33 \text{ kV}$$

5-9

由高-低压侧确定高压侧分接头，高-中压侧确定中压侧分接头。

(1) 高-低压侧分析：

最大负荷时： $S_{1\max} = 15 + j11 \text{ MVA}$ ，最小负荷时， $S_{1\min} = 8 + j6 \text{ MVA}$ 。

$$U_{4\max} = 110 - \frac{15 \times 2.71 + 11 \times 59.4}{110} = 103.69 \text{ kV}$$

$$U_{4\min} = 108 - \frac{8 \times 2.71 + 6 \times 59.4}{108} = 104.50 \text{ kV}$$

$$U'_{2\max} = 103.69 - \frac{10 \times 1.83 + 8 \times 38.3}{103.69} = 100.56 \text{ kV}$$

$$U'_{2\min} = 104.5 - \frac{5 \times 2.71 + 4 \times 59.4}{104.5} = 102.10 \text{ kV}$$

$$U'_{3\max} = 103.69 - \frac{1.6 \times 5 + 3 \times (-2.5)}{103.69} = 103.69 \text{ kV}, U'_{3\min} = 104.5 - \frac{1.6 \times 3 + 2 \times (-2.5)}{104.5} = 104.50 \text{ kV}$$

对于 T_3 : 低压侧为恒调压, 要求母线电压为 $1.02 \sim 1.05U_{3N}$, $U_{3N} = 6 \text{ kV}$.

$$\frac{U'_{3\max}}{1.02U_{3N}} = \frac{U_{t\max}}{6.6} \Rightarrow U_{t\max} = \frac{103.69 \times 6.6}{1.02 \times 6} = 111.82 \text{ kV}$$

$$\frac{U'_{3\min}}{1.05U_{3N}} = \frac{U_{t\min}}{6.6} \Rightarrow U_{t\min} = \frac{104.5 \times 6.6}{1.05 \times 6} = 109.48 \text{ kV}$$

$\Rightarrow U_t = 110.65 \text{ kV}$. 选择 110 kV , 校验此处略.

又对于 T_2 : 中压侧为顺调压, 要求母线为 $1.025 \sim 1.075U_{2N}$, $U_{2N} = 35 \text{ kV}$.

$$\therefore \frac{U'_{2\max}}{1.025U_{2N}} = \frac{110}{U_{t\max}} \Rightarrow U_{t\max} = \frac{110 \times 1.025 \times 35}{100.56} = 39.24 \text{ kV}$$

$$\frac{U'_{2\min}}{1.075U_{2N}} = \frac{110}{U_{t\min}} \Rightarrow U_{t\min} = \frac{110 \times 1.075 \times 35}{102.1} = 40.54 \text{ kV}$$

$$\Rightarrow U_t = 39.89 \text{ kV}, \quad k = \frac{39.89}{38.5} = 1.036 \Rightarrow \text{选} + 2.5\%.$$

$$\therefore U_t = 38.5 \times 1.025 = 39.4625 \text{ kV}$$

5-10

逆调压方式: 最大负荷时, $U_2 = 1.05U_{2N}$, 最小负荷时, $U_2 = U_{2N}$, $U_{2N} = 10 \text{ kV}$

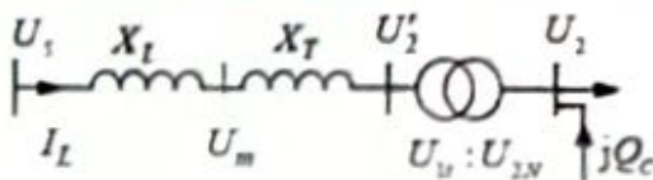


图 5.2: 题 5-10 电路图

(1) 先由最小负荷时确定 k : 最小负荷时, 令 $Q_C = 0$

$$I = \frac{U_s - U_{\min}}{X_L} = 0.54 \text{ A}, \quad U'_{2\min} = U_{1\min} - IX_T$$

$$= 107.5 - 0.54 \times 20 = 96.7 \text{ kV}$$

$$\therefore \frac{U'_{2\min}}{U_{2N}} = \frac{U_t}{11} \Rightarrow U_t = 11 \times \frac{96.7}{10} = 106.37 \text{ kV} \Rightarrow \text{选} 107.25 \text{ kV, 验算略}$$

(2) 再由最大负荷时确定 Q_C :

由大方式:

$$\left(U_s - \frac{P_{\max}R - (Q_{\max} - Q_C)X}{U_s} \right) \times \frac{U_{2N}}{U_{tr}} \geq 1.05U_N$$

$$Q_C \geq \left(1.05U_N \times \frac{U_{1t}}{U_{2N}} - U'_{2(\max)} \right) \times \frac{U_s}{X} \Rightarrow$$

$$Q_C \geq \left(1.05 \times 10 \times \frac{107.25}{11} - 84.1 \right) \times \frac{121}{45} = 49.139 \text{ Mvar}$$

最终结果:

$$\begin{cases} U_{1t} = 107.25 \text{ kV} \\ Q_C \geq 49.139 \text{ Mvar} \end{cases}$$

5-12

等微增量:

$$\frac{dF_1}{dP_{G1}} = 0.002P_{G1} + 0.2 = \lambda$$

$$\frac{dF_2}{dP_{G2}} = 0.004P_{G2} + 0.2 = \lambda$$

$$\Rightarrow P_{G1} = 2P_{G2}$$

$$\text{由 } P_{G1} + P_{G2} = 300 \text{ MW} \Rightarrow \begin{cases} P_{G1} = 200 \text{ MW} \\ P_{G2} = 100 \text{ MW} \end{cases}$$

分为负荷为 450MW 和 850MW 来讨论：

450 MW 时，火电机组 P_{G1} ，水电 P_{H1} ， $P_{G1} + P_{H1} = 450$ MW

850 MW 时，火电机组 P_{G2} ，水电 P_{H2} ， $P_{G2} + P_{H2} = 850$ MW

$$\begin{aligned} \text{等微增量: } & \begin{cases} 1 + 0.004P_{G1} = \mu(1 + 0.004P_{H1}) \\ 1 + 0.004P_{G2} = \mu(1 + 0.004P_{H2}) \end{cases} \\ & \begin{cases} 1 + 0.004(450 - P_{H1}) = \mu(1 + 0.004P_{H1}) \\ 1 + 0.004(850 - P_{H2}) = \mu(1 + 0.004P_{H2}) \end{cases} \\ \Rightarrow & \begin{cases} P_{H1} = \frac{2.8 - \mu}{0.004(\mu + 1)} \\ P_{H2} = \frac{4.4 - \mu}{0.004(\mu + 1)} \end{cases} \end{aligned}$$

日用水量：注意耗水率特性单位为 m^3/s

$$\begin{aligned} W &= 14 \times 3600(3 + P_{H1} + 0.002P_{H1}^2) + 10 \times 3600(3 + P_{H2} + 0.002P_{H2}^2) = 10^7 \\ \Rightarrow & 14 \times 36 \left(3 + \frac{2.8 - \mu}{0.004(\mu + 1)} + 0.002 \left(\frac{2.8 - \mu}{0.004(\mu + 1)} \right)^2 \right) \\ & + 10 \times 36 \left(3 + \frac{4.4 - \mu}{0.004(\mu + 1)} + 0.002 \left(\frac{4.4 - \mu}{0.004(\mu + 1)} \right)^2 \right) = 10^5 \\ & 1512 + 126000 \times \frac{2.8 - \mu}{\mu + 1} + 63000 \left(\frac{2.8 - \mu}{\mu + 1} \right)^2 + 1080 \\ & + 90000 \frac{4.4 - \mu}{\mu + 1} + 45000 \left(\frac{4.4 - \mu}{\mu + 1} \right)^2 = 10^5 \\ & 12600 \times \frac{2.8 - \mu}{\mu + 1} + 63000 \left(\frac{2.8 - \mu}{\mu + 1} \right)^2 + 1080 + 90000 \frac{4.4 - \mu}{\mu + 1} \\ & + 45000 \left(\frac{4.4 - \mu}{\mu + 1} \right)^2 = 97408 \end{aligned}$$

$$\frac{2.8(2.8 - \mu)}{\mu + 1} + 1.4 \left(\frac{2.8 - \mu}{\mu + 1} \right)^2 + \frac{2(4.4 - \mu)}{\mu + 1} + 1 \left(\frac{4.4 - \mu}{\mu + 1} \right)^2 = 2.1646$$

$$\frac{7.84 - 2.8\mu}{\mu + 1} + \frac{1.4(7.84 - 5.6\mu + \mu^2) + \mu^2 - 8.8\mu + 19.36}{(\mu + 1)^2} = 2.1646, \text{ 同乘 } (\mu + 1)^2$$

$$(16.64 - 4.8\mu)(\mu + 1) + 2.4\mu^2 - 16.64\mu + 30.336 = 2.1646(\mu + 1)^2$$

$$16.64\mu - 4.8\mu^2 + 16.64 - 4.8\mu + 2.4\mu^2 - 16.64\mu + 30.336$$

$$= 2.1646\mu^2 + 4.3292\mu + 2.1646$$

$$4.5646\mu^2 + 9.1292\mu - 44.8114 = 0$$

$$\mu_1 = 2.29, \mu_2 = -4.29 \text{ (舍)}, \text{ 代回即可}$$